

SREEANSH DASH

B.Tech Computer Science & Engineering (AI/ML) | VIT Chennai

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PROFESSIONAL SUMMARY

Third-year Computer Science student specializing in AI/ML with a 9.62 CGPA, seeking software, AI/ML and related internships. Hands-on developer of predictive models, multi-label diagnostic engines, computer vision pipelines, and intelligent edge systems. Proven competence in handling high-dimensional, imbalanced data, optimizing complex loss metrics, and scaling machine learning prototypes into functional software.

EDUCATION

Vellore Institute of Technology (VIT), Chennai

2024 – July 2028 (Expected)

B.Tech in Computer Science & Engineering (AI/ML)

CGPA: 9.62 / 10.0

Delhi Public School, Bhilai

Graduated 2024

CBSE Class XII

93.4%

TECHNICAL SKILLS

Languages Python, Java, C, C++, JavaScript, SQL

Machine Learning & Deep Learning Multi-Label Classification, Unsupervised Clustering, Hyperparameter Tuning, Cost-Sensitive Learning, Computer Vision, NLP

Developer Tools & Software Git, GitHub, REST APIs, Streamlit, VS Code, Cisco Packet Tracer, Figma

Frameworks & Libraries Scikit-Learn, XGBoost, LightGBM, AdaBoost, YOLOv8, OpenCV, NumPy, Pandas

Hardware / IoT Embedded Systems, Arduino, ESP32, Ultrasonic & IMU Sensors

PROJECTS

Multi-Label Clinical Diagnostic & Patient Stratification Engine

Machine Learning Project

Python · XGBoost · LightGBM · Scikit-Learn · Dimensionality Reduction

- Engineered an end-to-end predictive and unsupervised learning pipeline on a dataset of 99,992 patient records containing 109 clinical biomarkers.
- Implemented dimensionality reduction using Principal Component Analysis (PCA), retaining 55 components to capture 95% of global data variance, followed by K-Means clustering (K=5) to isolate distinct patient phenogroups.
- Formulated specialized label generation logic using localized anomaly detection (5th/95th percentiles) and mapped 70% structural divergence between cluster-specific and global clinical thresholds.
- Counteracted heavy class imbalance (ranging from 8.18% to 26.78% positive rates) by computing exact cost-sensitive `scale_pos_weight` metrics (up to 11.29) across 6 distinct diagnostic health labels.
- Evaluated and deployed a cost-weighted XGBoost ensemble model, achieving an overall 0.731 Macro AUC-ROC across multiple complex medical risk profiles

AI-Powered Grievance Redressal System

Hackathon Project

Python · NLP · Backend REST API · Streamlit

- Built a full-stack web application that uses **NLP-based automatic classification** to route citizen complaints to the correct department — eliminating manual triage entirely.
- Designed a status lifecycle engine to track complaints from submission through resolution, improving processing consistency and auditability.
- Delivered a functional prototype under hackathon time constraints, demonstrating end-to-end system design from database schema to UI.

Real-Time Automatic Number Plate Recognition (ANPR)

Personal Project

Python · YOLOv8 · OpenCV · Edge Deployment

- Trained and fine-tuned a **YOLOv8** object detection model for high-speed, accurate license plate localization and character recognition.
- Optimized the inference pipeline for edge deployment, prioritizing low-latency detection suitable for real-time traffic monitoring environments.
- Engineered a modular, scalable architecture that separates detection, OCR, and data-logging layers for easy extension.

Smart Walking Stick for the Visually Impaired

Course Project

C++ · ESP32 · Arduino · IMU · Ultrasonic Sensors

- Engineered an assistive IoT device using **ESP32**, ultrasonic sensors, and IMU-based fall detection to provide real-time hazard awareness and automated emergency alerts.
- Applied Ackermann steering geometry to the mechanical frame design, improving directional stability for users with varying mobility needs.
- Designed alert logic that triggers SMS/buzzer notifications on fall detection, enabling immediate caregiver response.
- **Innovation Impact:** Patent application currently under review.

EXPERIENCE & LEADERSHIP

Operations Lead — KalingaJyoti, VIT Chennai

July, 2025 – April 2026

- Managed a team of 20+ volunteers to plan and execute large-scale college events, overseeing logistics, scheduling, and cross-functional coordination.
- Developed communication and people-management skills by aligning diverse teams around shared goals under tight deadlines.

Open-Source Contributor — Y-SoC

March 2026- May 2026

- Actively contributing to community-driven open-source projects through code reviews, feature development, and collaborative problem-solving.

CERTIFICATIONS

Google UX Design Professional Certificate

Google / Coursera

- Completed an 8-course series covering user research, wireframing, low- and high-fidelity prototyping, and usability testing using Figma.

Getting Started with Cisco Packet Tracer

Cisco Networking Academy

- Verified proficiency in network simulation, device configuration, and topology design using Cisco Packet Tracer.

ADDITIONAL INFORMATION

Availability	Seeking to join Amazon ML summer school program and Seeking Summer 2027 internship, open to Bengaluru, Hyderabad, Bhubaneshwar.
Interests	Sustainable technology for social impact, competitive programming, IoT hardware hacking
Core Competencies	Data Structure & Algorithms, Competitive programming and Database Management.
Languages	English (Fluent), Hindi (Native), Odia (Native)